[redacted]

U.S. Citizen | [redacted] | [redacted].edu | linkedin.com/in/[redacted] | github.com/[redacted]

Education

[redacted]

Bachelor of Science in Computer Science, 3.85 GPA *May 2027*

- Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Competitive Programming, Programming in C, Programming in Java, Object-Oriented Programming

Experience

NSF Data Science & AI Researcher January 2025 - Present

[redacted]

- Investigated semiconductor reliability with the **NSF** by analyzing **4,500+** X-ray tomography images
- Built a **Python/OpenCV** image processing pipeline to identify defects in solder balls with **97%** accuracy
- Automated defect tracking, a **200x** improvement over hand labeling speed to generate data and visualizations
- Performed analyses on **10k** lines of data and **6 datasets** with **Pandas** to discover previously unknown correlations

Software Engineering Intern August 2024 – May 2025

[redacted]

- Analyzed over **800k** flight telemetry data points by developing an interactive data visualization app with **Panel**
- Improved performance by **10x** by optimizing data analysis to increase accessibility for engineers **(5.6s to 0.56s)**
- Engineered machine learning algorithms trained on **60+** datasets to simulate test flights instead of physical testing
- Coordinated **Agile and Scrum** methodologies like sprints and retrospectives within a **10+** member team

Projects

[redacted] | *Next.js, React, Python, FastAPI, Llama LLM*

- Built fact-checking tool at [redacted] **(1st of 400+)**, winning the [redacted] **hackathon** in the US
- Developed full-stack system using **Next.js** frontend, **Python/FastAPI** backend and DuckDuckGo web scraping
- Produced fact verification tested on **10+ hours** of video with web scraping and locally hosted Llama LLM
- Created a Chrome extension with **RESTful APIs** integrated with webapp for **95%** accurate fact-checking

[redacted] **Website** | *Framer, HTML, CSS*

- Designed a landing page serving over **5k monthly users** for [redacted] an [redacted] client
- Promoted app to reach **200k total downloads** and span **40+ countries** across the app's lifetime
- Implemented interactive product showcase, FAQ, and streamlined download process to improve user experience
- Integrated analytics tools to track user engagement and enhance website performance

Procedural Terrain Generation Engine | *Unity, C#, React.js*

- Developed terrain generation engine in Unity generating chunks in **22ms** avg at over **300 FPS**
- Combined multiple **Perlin noise** layers to introduce large and small features like mountains and rocks
- Optimized performance with LOD and meshes, reducing rendering to **400 triangles** for real-time generation
- Created a companion website using **React.js** to showcase terrain and allow users to generate **65k** custom seeds

Leadership

VEX Robotics Competition

Team Captain/Programming Lead

- **2x National** [redacted], **1st/2nd** at [redacted] State Championship, **4x Tournament Wins**
- Coordinated and taught version control for **20+ programmers** using **Git** and **GitHub** for seamless collaboration
- Designed and tuned **PID controllers** for odometry to navigate during **15 sec** autonomous period
- Developed vision auto-aiming system in **C++** with closed-loop flywheel RPM to score from anywhere on the field

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, C#, C, SQL/MySQL, R, HTML/CSS, LaTeX, Rust
Frameworks & Technologies: Pandas, NumPy, React, Next.js, Node.js, OpenCV, PyTorch, Tensorflow, Springboot
Developer Tools: Git, Linux, VSCode, Unity, Docker, Kubernetes, Vim/Neovim, Azure, AWS, Google Firebase